Experiment No. 6

Title

Managing User and Groups in Linux

Aim

The problem is to create, modify, and manage users and groups in Linux, understand file ownership and group permissions, and practice sharing files securely between users and groups.

Theory

Users in Linux

Each user in Linux has:

- A unique username
- A user ID (UID)
- A home directory
- A default shell

Linux supports different types of users:

- Root user – Superuser with full system privileges
- Normal users – Regular users with limited privileges

Groups in Linux

Groups are collections of users that simplify permission management. Each file or directory belongs to:

- One owner
- One group

File Ownership

Ownership determines who can read, write, or execute files. The commands chown and chgrp are used to modify ownership.

Sticky Bit

The sticky bit is a special permission that prevents users from deleting files owned by others in a shared directory. Example: chmod +t directory_name

Commands Used

Commands	Description
useradd	Create a new user
passwd	Set or change user password
usermod	Modify user account
userdel	Delete a user
groupadd	Create a new group
groupdel	Delete a group
groups	Display groups of a user
chown	Change file owner

Procedure / Steps

1) Log in to the Linux system.
2) Open the terminal.
3) Create a new user:

- sudo useradd testuser
- sudo passwd testuser

4) Verify the user:

- id testuser

5) Create a new group:

- sudo groupadd testgroup

6) Add user to the group:

- sudo usermod -aG testgroup testuser

7) Check user group membership:

- groups testuser

8) Create a shared directory:

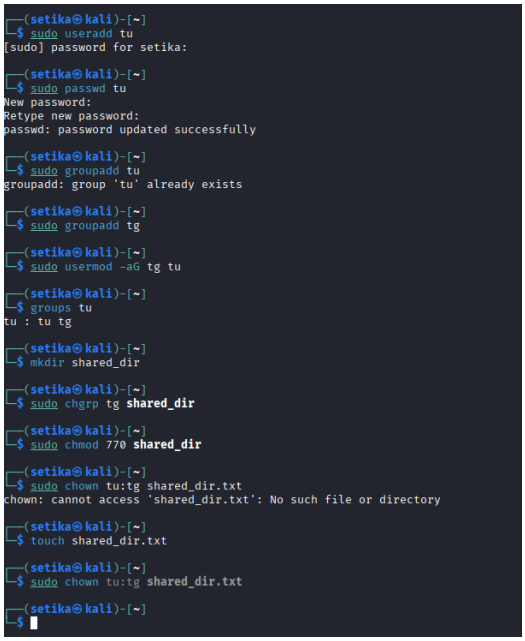
- mkdir shared_dir
- sudo chgrp testgroup shared_dir
- sudo chmod 770 shared_dir

9) Apply sticky bit:

- sudo chmod +t shared_dir

10) Change file ownership:

- sudo chown testuser:testgroup shared_file.txt



Outcomes

- ✓ New users and groups created successfully.
- ✓ Users added to groups correctly.
- ✓ File and directory ownership modified.
- ✓ Shared directory protected using sticky bit.

Result

- ❖ Linux user and group management operations were successfully performed. The student learned to control access to system resources using users, groups, permissions, and special access bits.